

Downloads

Model and related tools

For questions about the code, please make sure to check the FAQs before contacting us. The SMPL model is available for non-commercial research, education, and artistic purposes only. By downloading the model you agree to the [license terms](#).

The model is also available for commercial purposes by contacting sales@meshcapade.com.

If you are unsure whether your use counts as non-commercial please contact smpl@tuebingen.mpg.de before downloading the model.

Overview:

SMPL is an easy-to-use, realistic, model of the of the human body that is useful for animation and computer vision.

SMPL lets you create realistic human body shapes with realistic deformations.

Available here is a version of SMPL that is similar to that described in [this paper](#).

SMPL for Python Users

Included in this download:

- Model files
 - PKL file format
 - separate male and female model files



- Code
 - Functions to load / save SMPL models
 - 'Hello World' sample scripts

System Requirements

Operating system: macOS, Linux

Python 2.7 Dependencies

- [Numpy & Scipy](#)
- [Chumpy](#)
- [OpenCV](#)

Download

[Download version 1.0.0 for Python 2.7 \(female/male. 10 shape PCs\)](#)

[Download version 1.1.0 for Python 2.7 \(female/male/neutral, 300 shape PCs\)](#)

[Download UV map in OBJ format](#)

Julia Models

[Male \(43 MB\)](#) [Female \(43 MB\)](#) [Neutral \(43 MB\)](#)

DMPLs for AMASS

- 8 DMPL components to control soft tissue dynamics
- Available for male, female, and neutral models
- Includes .npz numpy arrays
- For an example usage refer to project [AMASS](#)
- If you use DMPLs, in addition to SMPL, please also cite the reference for [Dyna](#)

Download

[Download DMPLs compatible with SMPL](#)

Sample Animated Bodies

These are FBX files that you can load and play in most graphics software



[Download sample animated bodies FBX files](#)

SMPL for Maya

Included in this download:

- Male and female models (FBX format)
 - Female model:
 - basicModel_f_lbs_10_207_0_v1.0.2.fbx
 - MD5: 81a57177f8cbf584f8d6379742c99e2e
 - Also referred to as: SMPL_female_010_207.fbx
 - Male model:
 - basicModel_m_lbs_10_207_0_v1.0.2.fbx
 - MD5: 67542f27aa4bc6b8d12da28fdce51539
 - Also referred to as: SMPL_male_010_207.fbx
 - Default mesh. The average SMPL body.
 - Rigged skeleton, using LBS
 - 10 shape blendshapes

- 207 corrective pose blendshapes
- UV map
- Script and data to use SMPL in Maya
 - If you want to edit body shape or repose the mesh, you need to run the script.
- README.txt for installation and license instructions

Dependencies

Numpy is required for running this script. Numpy is a Python module that can be installed following the instructions given [here](#) or [here](#)

Please make sure you have numpy installed on your computer and accessible through Maya's Python. We are working towards removing this dependency.

Download

[Release notes for version 1.0.2](#)

[Download version 1.0.2 for Maya](#)

[Download SMPL-Maya loader by BodyLabs](#)

Tutorial videos

Tutorial videos for Maya can be found in the [Videos](#) section.

SMPL for Unity

Included in this download:

- Male and female models (Unity-compatible FBX format)
 - Default mesh. The average SMPL body.
 - Rigged skeleton, using LBS
 - 10 shape blendshapes

- 207 corrective pose blendshapes
- UV map
- Scripts and sample Unity project to help you get started
 - If you want to edit body shape or repose the mesh, you need the scripts.
- README.txt for installation and license instructions
- How-to_SMPLinUnity.pdf for step-by-step guide for using SMPL in Unity

System Requirements

Unity 5+ (tested on macOS Sierra and Windows 10)

Download

[Download version 1.0.0 for Unity](#)

How to use SMPL in Unity

[Download tutorial \(PDF\)](#)

SMPL for Blender

This Blender add-on allows you to add SMPL skinned meshes to your current Blender scene. Each imported SMPL mesh consists of a shape specific rig, as well as shape keys (blend shapes) for shape and pose correctives.

- Requirements: Blender 2.80+
- Additional dependencies: None
- Used SMPL model: SMPL v1.0 with 10 shape components

Installation instructions

[Download SMPL Blender Add-on \(20210611, 34MB\)](#)

SMPL for Unreal Plugin (USMPL)

The USMPL plugin allows a user to put the SMPL model in Unreal Engine and control its pose using ROS in real-time. The plugin is available at <https://github.com/robot-perception-group/USMPL>. Following are the gender-specific models for the USMPL plugin.

`basicModel_f_lbs_10_207_0_v1.0.0.npz`

`basicmodel_m_lbs_10_207_0_v1.0.0.npz`

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